

FH fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FH gene fusions from the msk_impact_50k_2026 study, covering 16 fusion events. 1/16 fusions (6.2%) were in-frame. 2/16 fusions (12.5%) retained the PF00206 domain. Domain retention was tested with Fisher's exact test ($p=0.133333$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for FH.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 16 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was DGKH, observed in 5 events.

Domain retention

PF00206 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 1/1 (100.0%) of in-frame fusions retained the domain, $p=0.133333$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.287129.

Domain disruption

Domain-disruption analysis was skipped: FH has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 15 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 45 aa (permutation-corrected $p=0.0689311$, not statistically significant at $\alpha=0.05$). This is -13 aa from the nearest configured domain boundary at 58 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=2$) and "not_retained" ($n=13$). For Frame_status, retained: 1/1 (100.0%, 95% CI 20.7%-100.0%); not_retained: 0/5 (0.0%, 95% CI 0.0%-43.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 12 vs 29.3077, $p=0.0682804$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: FH has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RYR2	1	0.0625	0.05419233758836865	None	1.0	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27612678359599413	1
unknown	1	0.0625	0.05419233758836865	None	1.0	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27612678359599413	2
PDE1C	1	0.0625	0.05419233758836865	None	0.9976851851851852	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27569275581821634	3
DGKH	5	0.3125	0.05419233758836865	None	0.20416666666666666	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22065803359599415	4
NOTCH1	1	0.0625	0.05419233758836865	None	0.6736111111111112	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.21492886692932747	5
RGS7	2	0.125	0.05419233758836865	None	0.3912037037037037	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.18541497804043855	6
FMN2	1	0.0625	0.05419233758836865	None	0.3055555555555556	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.14591845026266081	7
ARID4B	1	0.0625	0.05419233758836865	None	0.24768518518518523	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.13506775581821637	8
TRMT1L	1	0.0625	0.05419233758836865	None	None	0.3860334247970314	['confidence_certainty', 'domain_retention', 'recurrence']	0.10907911827199278	9
MAFTRR	1	0.0625	0.05419233758836865	None	0.02314814814814814	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.0929670613737719	10
SLC35F3	1	0.0625	0.05419233758836865	None	0.0	0.3860334247970314	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.08862678359599413	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
44	0	2	1	12	0.0	1.0	-0.0
45	2	0	1	12	None	0.02857142857142857	1.5440680443502757
186	2	0	2	11	None	0.05714285714285714	1.2430380486862944
246	2	0	3	10	None	0.09523809523809523	1.021189299069938
345	2	0	4	9	None	0.14285714285714285	0.8450980400142568
370	2	0	10	3	None	1.0	-0.0
464	2	0	11	2	None	1.0	-0.0
467	2	0	12	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

1	1
0	13

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Lyase	15	2	9	4	0.45713623725671915

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	8	16	0.5

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ARID4B	1
DGKH	5

Partner_gene	Event_count
FMN2	1
MAFTRR	1
NOTCH1	1
PDE1C	1
RGS7	2
RYR2	1
SLC35F3	1
TRMT1L	1
unknown	1

Per-event results (results.tsv)

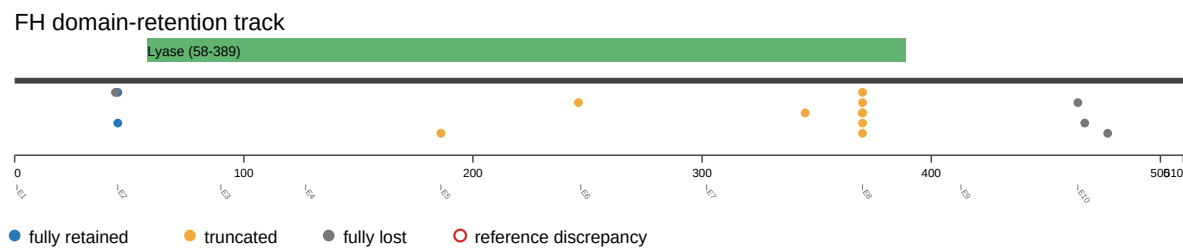
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EVT-P-004 8392-T01-I M6-1	P-0048 392-T0 1-IM6			unknown	unknown	three_prime	241680 684	2	45	True	retained	1.0	False	Lyase			FH (NM_000143) rearrangement: c.133-68:F H_chr1:g.240716629inv Transcript Fusion {MIR1273E:FH}	NA
EVT-P-006 2340-T01-I M7-5	P-0062 340-T0 1-IM7		ARID4B -FH	ARID4B	out-of-frame	three_prime	241665 896	8	370	True	disrupted	0.0602409 638554216 86	True			Lyase	ARID4B (NM_016374) - FH (NM_000143) rearrangement: c.1841+52 45:ARID4B_c.1109-26:FH dup Protein Fusion: out of frame {ARID4B:FH}	NA
EVT-P-001 8108-T01-I M6-7	P-0018 108-T0 1-IM6		DGKH- FH	DGKH	unknown	five_prime	241667 339	7	370	True	disrupted	0.9427710 843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q4 3;q14.11)(chr1:g.2416673 40::chr13:g.42830153) Transcript Fusion {FH:DGKH}	NA
EVT-P-001 8108-T02-I M6-8	P-0018 108-T0 2-IM6		DGKH- FH	DGKH	unknown	five_prime	241667 340	7	370	True	disrupted	0.9427710 843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q4 3;q14.11)(chr1:g.2416673 40::chr13:g.42830153) Transcript Fusion {FH:DGKH}	NA
EVT-P-006 4901-T01-I M7-9	P-0064 901-T0 1-IM7		DGKH- FH	DGKH	unknown	five_prime	241667 338	7	370	True	disrupted	0.9427710 843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q4 3;q14.11)(chr1:g.2416673 38::chr13:g.42830159) Transcript Fusion {FH:DGKH}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 9293-T01-I M7-10	P-0059 293-T0 1-IM7		DGKH-FH	DGKH	unknown	five_prime	241667341	7	370	True	disrupted	0.9427710 843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667341::chr13:g.42830159) Protein Fusion: mid-exon {FH:DGKH}	NA
EVT-P-005 9435-T01-I M7-11	P-0059 435-T0 1-IM7		DGKH-FH	DGKH	unknown	three_prime	241661272	10	464	True	lost	0.0	False		Lyase		FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241661272::chr13:g.42830525) Transcript Fusion {DGKH:FH}	NA
EVT-P-006 6922-T02-I M7-20	P-0066 922-T0 2-IM7		FMN2-FH	FMN2	unknown	five_prime	241667416	7	345	False	disrupted	0.8674698 795180723	True			Lyase	FH (NM_000143) rearrangement: c.1034:FH_chr1:g.240469632del Antisense Fusion	NA
EVT-P-006 0877-T01-I M7-21	P-0060 877-T0 1-IM7		MAFTR-FH	MAFTR	unknown	three_prime	241661260	10	467	False	lost	0.0	False		Lyase		FH (NM_000143) rearrangement: t(1;16)(q43;q23.2)(chr1:g.241661260::chr16:g.79754421) Protein Fusion: mid-exon {LOC102467146:FH}	NA
EVT-P-004 0956-T01-I M6-22	P-0040 956-T0 1-IM6		NOTCH1-FH	NOTCH1	out-of-frame	three_prime	241672198	5	186	True	disrupted	0.6144578 313253012	True			Lyase	NOTCH1 (NM_017617) - FH (NM_000143) rearrangement: t(1;9)(q43;q34.3)(chr1:g.241672198::chr9:g.139417940) Protein Fusion: out of frame {NOTCH1:FH}	NA
EVT-P-000 0532-T02-I M5-25	P-0000 532-T0 2-IM5		PDE1C-FH	PDE1C	out-of-frame	five_prime	241682866	1	44	True	lost	0.0	False		Lyase		FH (NM_000143) - PDE1C (NM_001191058) Rearrangement : t(1;7)(1q43;p14.3)(chr1:g.241682866::chr7:g.31926696) Protein fusion: out of frame {FH-PDE1C}	NA
EVT-P-003 2745-T01-I M6-26	P-0032 745-T0 1-IM6		RGS7-FH	RGS7	out-of-frame	five_prime	241671722	5	246	True	disrupted	0.5692771 084337349	True			Lyase	FH (NM_000143) - RGS7 (NM_002924) rearrangement: c.738+181:FH_c.386-8492:RGS7del Protein Fusion: out of frame {FH:RGS7}	NA
EVT-P-000 4918-T02-I M6-27	P-0004 918-T0 2-IM6		RGS7-FH	RGS7	out-of-frame	three_prime	241666009	8	370	True	disrupted	0.0602409 638554216 86	True			Lyase	RGS7 (NM_002924) - FH (NM_000143) rearrangement: c.79-95591:RGS7_c.1109-139:FHdup Protein Fusion: out of frame {RGS7:FH}	NA

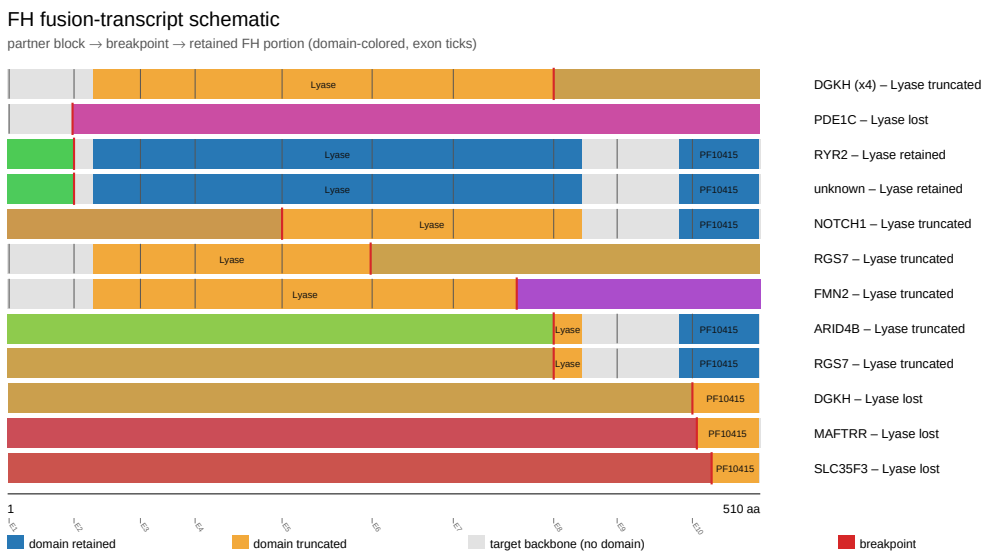
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EVT-P-0015961-T03-IM6-28	P-0015961-T03-IM6		RYR2-FH	RYR2	in-frame	three_prime	241680785	2	45	True	retained	1.0	False	Lyase			RYR2 (NM_001035) - FH (NM_000143) rearrangement: c.48+56557:RYR2_c.133-169:FHinv Protein Fusion: in frame {RYR2:FH}	NA
EVT-P-0051358-T01-IM6-29	P-0051358-T01-IM6		SLC35F3-FH	SLC35F3	unknown	three_prime	241661231	10	477	False	lost	0.0	False		Lyase		SLC35F3 (NM_173508) - FH (NM_000143) rearrangement: c.283+157829:SLC35F3_c.1430:FHinv Protein Fusion: mid-exon {SLC35F3:FH}	NA

Figures

domain retention outliers



fusion schematic



reference comparison

Reference vs msk_impact_50k_2026

